



llama-3.1-8b-instruct-fast

Text Generation • Meta

@cf/meta/llama-3.1-8b-instruct-fast

[Fast version] The Meta Llama 3.1 collection of multilingual large language models (LLMs) is a collection of pretrained and instruction tuned generative models. The Llama 3.1 instruction tuned text only models are optimized for multilingual dialogue use cases and outperform many of the available open source and closed chat models on common industry benchmarks.

Model Info

Context Window [↗](#)

128,000 tokens

Model Info

Terms and License [link ↗](#)

Playground

Try out this model with Workers AI LLM Playground. It does not require any setup or authentication and an instant way to preview and test a model directly in the browser.

Launch the LLM Playground

Usage

✓ Worker - Streaming

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const stream = await  
env.AI.run("@cf/meta/llama-3.1-8b-instruct-  
fast", {  
      messages,  
      stream: true,  
    });
```

```
return new Response(stream, {  
  headers: { "content-type":  
"text/event-stream" },  
  });  
},  
} satisfies ExportedHandler<Env>;
```

✓ Worker

TypeScript

```
export interface Env {  
  AI: Ai;  
}  
  
export default {  
  async fetch(request, env):  
    Promise<Response> {  
  
    const messages = [  
      { role: "system", content: "You are a  
friendly assistant" },  
      {  
        role: "user",  
        content: "What is the origin of the  
phrase Hello, World",  
      },  
    ];  
  
    const response = await  
env.AI.run("@cf/meta/llama-3.1-8b-instruct-  
fast", { messages });  
  
    return Response.json(response);  
  },  
} satisfies ExportedHandler<Env>;
```

▼ Python

Python

```
import os
import requests

ACCOUNT_ID = "your-account-id"
AUTH_TOKEN = os.environ.get("CLOUDFLARE_AUTH_TOKEN")

prompt = "Tell me all about PEP-8"
response = requests.post(
    f"https://api.cloudflare.com/client/v4/accounts/{ACCOUNT_ID}/llama-3.1-8b-instruct-fast",
    headers={"Authorization": f"Bearer {AUTH_TOKEN}"},
    json={
        "messages": [
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt}
        ]
    }
)
result = response.json()
print(result)
```

▼ curl

```
curl
```

```
https://api.cloudflare.com/client/v4/accounts/3.1-8b-instruct-fast \
-X POST \
-H "Authorization: Bearer $CLOUDFLARE_AUTH_TOKEN" \
-d '{ "messages": [{ "role": "system", "content": "You are a helpful assistant." }, { "role": "user", "content": "Why is pizza so good" } ] }'
```

① OpenAI compatible endpoints

Workers AI also supports OpenAI compatible API endpoints for `/v1/chat/completions` and `/v1/embeddings`. For more details, refer to [Configurations](#).

Parameters

* indicates a required field

Input

- **0** **object**

- frequency_penalty **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- image **one of**

- **0** **array**

An array of integers that represent the image data constrained to 8-bit unsigned integer values

- items **number**

A value between 0 and 255

- **1** **string**

Binary string representing the image contents.

- lora **string**

Name of the LoRA (Low-Rank Adaptation) model to fine-tune the base model.

- `max_tokens` **integer** default 256

The maximum number of tokens to generate in the response.

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- `prompt` **string** required min 1 max 131072

The input text prompt for the model to generate a response.

- `raw` **boolean**

If true, a chat template is not applied and you must adhere to the specific model's expected formatting.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- `seed` **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **stream** **boolean**

If true, the response will be streamed back incrementally using SSE, Server Sent Events.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **top_k** **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- **top_p** **number** min 0 max 2

Adjusts the creativity of the AI's responses by controlling how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

- **1** **object**

- **frequency_penalty** **number** min 0 max 2

Decreases the likelihood of the model repeating the same lines verbatim.

- functions **array**

- items **object**

- code **string** required

- name **string** required

- image **one of**

- 0 **array**

- An array of integers that represent the image data constrained to 8-bit unsigned integer values

- items **number**

- A value between 0 and 255

- 1 **string**

- Binary string representing the image contents.

- max_tokens **integer** default 256

The maximum number of tokens to generate in the response.

- `messages` **array** required

An array of message objects representing the conversation history.

- `items` **object**

- `content` **string** required max 131072

The content of the message as a string.

- `role` **string** required

The role of the message sender (e.g., 'user', 'assistant', 'system', 'tool').

- `presence_penalty` **number** min 0 max 2

Increases the likelihood of the model introducing new topics.

- `repetition_penalty` **number** min 0 max 2

Penalty for repeated tokens; higher values discourage repetition.

- **seed** **integer** min 1 max 9999999999

Random seed for reproducibility of the generation.

- **stream** **boolean**

If true, the response will be streamed back incrementally.

- **temperature** **number** default 0.6 min 0 max 5

Controls the randomness of the output; higher values produce more random results.

- **tools** **array**

A list of tools available for the assistant to use.

- **items** **one of**

- **0** **object**

- **description** **string** required

A brief description of what the tool does.

- **name** **string** required

The name of the tool. More descriptive the better.

- **parameters** **object** required

Schema defining the parameters accepted by the tool.

- **properties** **object** required

Definitions of each parameter.

- **additionalProperties** **object**

- **description** **string** required

A description of the expected parameter.

- **type** **string** required

The data type of the parameter.

- **required** **array**

List of required parameter names.

- **items** **string**

- **type** **string** required

The type of the parameters object (usually 'object').

■ 1 **object**

■ function **object** required

Details of the function tool.

■ description **string** required

A brief description of what the function does.

■ name **string** required

The name of the function.

■ parameters **object** required

Schema defining the parameters accepted by the function.

■ properties **object** required

Definitions of each parameter.

■ additionalProperties **object**

■ description **string** required

A description of the expected parameter.

- type **string** required

The data type of the parameter.

- required **array**

List of required parameter names.

- items **string**

- type **string** required

The type of the parameters object (usually 'object').

- type **string** required

Specifies the type of tool (e.g., 'function').

- top_k **integer** min 1 max 50

Limits the AI to choose from the top 'k' most probable words. Lower values make responses more focused; higher values introduce more variety and potential surprises.

- top_p **number** min 0 max 2

Controls the creativity of the AI's responses by adjusting how many possible words it considers. Lower values make outputs more predictable; higher values allow for more varied and creative responses.

Output

- **0** **object**

- **response** **string**

The generated text response from the model

- **tool_calls** **array**

An array of tool calls requests made during the response generation

- **items** **object**

- **arguments** **object**

The arguments passed to be passed to the tool call request

■ name **string**

The name of the tool to be called

► 1 **string**

API Schemas

The following schemas are based on JSON Schema

Input

Output

```
{
  "oneOf": [
    {
      "properties": {
        "frequency_penalty": {
          "description": "Decreases
the likelihood of the model repeating the same
lines verbatim.",
          "maximum": 2,
          "minimum": 0,
          "type": "number"
        },
        "image": {
          "oneOf": [
            {
              "description": "An
array of integers that represent the image
data constrained to 8-bit unsigned integer
values",
              "items": {
                "description":
"A value between 0 and 255",
                "type":
"number"
```

```
        },
        "type": "array"
      },
      {
        "description":
"Binary string representing the image
contents.",
        "format":
"binary",
        "type": "string"
      }
    ]
  },
  "lora": {
    "description": "Name of
the LoRA (Low-Rank Adaptation) model to fine-
tune the base model.",
    "type": "string"
  },
  "max_tokens": {
    "default": 256,
    "description": "The
maximum number of tokens to generate in the
response.",
    "type": "integer"
  }
}
```



```
    },  
    "presence_penalty": {  
      "description": "Increases  
the likelihood of the model introducing new  
topics.",  
      "maximum": 2,  
      "minimum": 0,  
      "type": "number"  
    },  
    "prompt": {  
      "description": "The input  
text prompt for the model to generate a  
response.",  
      "maxLength": 131072,  
      "minLength": 1,  
      "type": "string"  
    },  
    "raw": {  
      "default": false,  
      "description": "If true, a  
chat template is not applied and you must  
adhere to the specific model's expected  
formatting.",  
      "type": "boolean"  
    },  
  },
```

```
"repetition_penalty": {  
  "description": "Penalty  
for repeated tokens; higher values discourage  
repetition.",  
  "maximum": 2,  
  "minimum": 0,  
  "type": "number"  
},  
"seed": {  
  "description": "Random  
seed for reproducibility of the generation.",  
  "maximum": 9999999999,  
  "minimum": 1,  
  "type": "integer"  
},  
"stream": {  
  "default": false,  
  "description": "If true,  
the response will be streamed back  
incrementally using SSE, Server Sent Events.",  
  "type": "boolean"  
},  
"temperature": {  
  "default": 0.6,
```

```
      "description": "Controls  
the randomness of the output; higher values  
produce more random results.",  
      "maximum": 5,  
      "minimum": 0,  
      "type": "number"  
    },  
    "top_k": {  
      "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more  
focused; higher values introduce more variety  
and potential surprises.",  
      "maximum": 50,  
      "minimum": 1,  
      "type": "integer"  
    },  
    "top_p": {  
      "description": "Adjusts  
the creativity of the AI's responses by  
controlling how many possible words it  
considers. Lower values make outputs more  
predictable; higher values allow for more  
varied and creative responses.",  
      "maximum": 2,
```

```
        "minimum": 0,
        "type": "number"
    },
    },
    "required": [
        "prompt"
    ],
    "title": "Prompt"
},
{
    "properties": {
        "frequency_penalty": {
            "description": "Decreases
the likelihood of the model repeating the same
lines verbatim.",
            "maximum": 2,
            "minimum": 0,
            "type": "number"
        },
        "functions": {
            "items": {
                "properties": {
                    "code": {
                        "type":
"string"
```

```
    },  
    "name": {  
      "type":  
"string"  
    }  
  },  
  "required": [  
    "name",  
    "code"  
  ],  
  "type": "object"  
},  
"type": "array"  
},  
"image": {  
  "oneOf": [  
    {  
      "description": "An  
array of integers that represent the image  
data constrained to 8-bit unsigned integer  
values",  
      "items": {  
        "description":  
"A value between 0 and 255",
```

```
        "type":  
        "number"  
    },  
    "type": "array"  
},  
{  
    "description":  
    "Binary string representing the image  
contents.",  
    "format":  
    "binary",  
    "type": "string"  
}  
]  
,  
"max_tokens": {  
    "default": 256,  
    "description": "The  
maximum number of tokens to generate in the  
response.",  
    "type": "integer"  
},  
"messages": {  
    "description": "An array  
of message objects representing the
```

```
conversation history.",
    "items": {
        "properties": {
            "content": {
                "description":
"The content of the message as a string.",
                "maxLength":
131072,
                "type":
"string"
            },
            "role": {
                "description":
"The role of the message sender (e.g., 'user',
'assistant', 'system', 'tool').",
                "type":
"string"
            }
        },
        "required": [
            "role",
            "content"
        ],
        "type": "object"
    },
```

```
      "type": "array"
    },
    "presence_penalty": {
      "description": "Increases
the likelihood of the model introducing new
topics.",
      "maximum": 2,
      "minimum": 0,
      "type": "number"
    },
    "repetition_penalty": {
      "description": "Penalty
for repeated tokens; higher values discourage
repetition.",
      "maximum": 2,
      "minimum": 0,
      "type": "number"
    },
    "seed": {
      "description": "Random
seed for reproducibility of the generation.",
      "maximum": 9999999999,
      "minimum": 1,
      "type": "integer"
    },
  },
```



```
"stream": {  
  "default": false,  
  "description": "If true,  
the response will be streamed back  
incrementally.",  
  "type": "boolean"  
},  
"temperature": {  
  "default": 0.6,  
  "description": "Controls  
the randomness of the output; higher values  
produce more random results.",  
  "maximum": 5,  
  "minimum": 0,  
  "type": "number"  
},  
"tools": {  
  "description": "A list of  
tools available for the assistant to use.",  
  "items": {  
    "oneOf": [  
      {  
        "properties":  
{
```

```
"description": {
```

```
"description": "A brief description of what  
the tool does.",
```

```
"type": "string"
```

```
},
```

```
"name": {
```

```
"description": "The name of the tool. More  
descriptive the better.",
```

```
"type": "string"
```

```
},
```

```
"parameters": {
```

```
"description": "Schema defining the parameters  
accepted by the tool.",
```

```
"properties": {
```

```
"properties": {
```

```
"additionalProperties": {  
  
  "properties": {  
  
    "description": {  
  
      "description": "A description of the expected  
parameter.",  
  
      "type": "string"  
  
    },  
  
    "type": {  
  
      "description": "The data type of the  
parameter.",  
  
      "type": "string"  
  
    }  
  
  },  
  
}
```

```
"required": [  
  
  "type",  
  
  "description"  
  
],  
  
"type": "object"  
  
},  
  
"description": "Definitions of each  
parameter.",  
  
"type": "object"  
  
},  
  
"required": {  
  
  "description": "List of required parameter  
names.",  
  
  "items": {
```

```
"type": "string"
```

```
},
```

```
"type": "array"
```

```
},
```

```
"type": {
```

```
"description": "The type of the parameters  
object (usually 'object').",
```

```
"type": "string"
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"properties"
```

```
],
```

```
"type": "object"
```

```
        }  
    },  
    "required": [  
        "name",  
  
        "description",  
  
        "parameters"  
    ],  
},  
{  
    "properties":  
{  
  
    "function": {  
  
        "description": "Details of the function  
tool.",  
  
        "properties": {  
  
            "description": {  
  
                "description": "A brief description of what  
the function does.",
```

```
"type": "string"
},

"name": {

"description": "The name of the function.",

"type": "string"
},

"parameters": {

"description": "Schema defining the parameters
accepted by the function.",

"properties": {

"properties": {

"additionalProperties": {

"properties": {

"description": {
```

```
"description": "A description of the expected  
parameter.",
```

```
"type": "string"
```

```
},
```

```
"type": {
```

```
"description": "The data type of the  
parameter.",
```

```
"type": "string"
```

```
}
```

```
},
```

```
"required": [
```

```
"type",
```

```
"description"
```



```
],  
  
"type": "object"  
  
},  
  
"description": "Definitions of each  
parameter.",  
  
"type": "object"  
  
},  
  
"required": {  
  
"description": "List of required parameter  
names.",  
  
"items": {  
  
"type": "string"  
  
},  
  
}
```

```
"type": "array"

},

"type": {

"description": "The type of the parameters
object (usually 'object').",

"type": "string"

},

"required": [

"type",

"properties"

],

"type": "object"
```

```
    },  
  },  
  
  "required": [  
  
    "name",  
  
    "description",  
  
    "parameters"  
  ],  
  
  "type": "object"  
},  
"type": {  
  
  "description": "Specifies the type of tool  
(e.g., 'function').",  
  
  "type": "string"  
}  
},  
"required": [  
  "type",  
  "function"
```

```
    ]  
  }  
],  
  "type": "object"  
},  
  "type": "array"  
},  
  "top_k": {  
    "description": "Limits the  
AI to choose from the top 'k' most probable  
words. Lower values make responses more  
focused; higher values introduce more variety  
and potential surprises.",  
    "maximum": 50,  
    "minimum": 1,  
    "type": "integer"  
},  
  "top_p": {  
    "description": "Controls  
the creativity of the AI's responses by  
adjusting how many possible words it  
considers. Lower values make outputs more  
predictable; higher values allow for more  
varied and creative responses.",  
    "maximum": 2,
```

```
        "minimum": 0,  
        "type": "number"  
      }  
    },  
    "required": [  
      "messages"  
    ],  
    "title": "Messages"  
  }  
],  
"type": "object"  
}
```

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